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EXPERIMENT NO. 07

Aim

To implement monitoring using Prometheus / Grafana / Nagios.

Theory

Continuous monitoring is a process to detect, report and respond to attacks or issues which occur in infrastructure. Once an application is deployed into a server, continuous monitoring helps maintain the infrastructure and respond appropriately.

Nagios is used to detect network or server issues, monitor infrastructure and business processes, troubleshoot server performance issues, plan infrastructure upgrades, and maintain service security and availability.

NRPE (Nagios Remote Plugin Executor) executes monitoring plugins on remote systems. For Windows hosts, NSClient++ acts as the remote agent and can execute local checks such as disk, CPU, memory and service checks.

Requirements / Environment Used

- Windows host
- Docker Desktop
- Nagios Core 4.5.12 running in Docker
- NSClient++ 0.6.13 x64
- NRPE communication
- Nagios Web Interface

Procedure

1. Verify Docker: Docker was verified from PowerShell using docker --version and docker version.

2. Pull and run Nagios: The Nagios Docker image jasonrivers/nagios:latest was pulled and the container nagios4 was started with port 8080 mapped to port 80.

3. Configure Nagios administrator: The nagiosadmin account password was configured and the Nagios Web Interface was opened at http://localhost:8080/nagios/.

4. Install NSClient++: NSClient++ 0.6.13 x64 was installed on the Windows machine and the nscp service was verified as running.

5. Configure NSClient++: NRPEServer, CheckDisk and CheckSystem were enabled. The Docker gateway address was added to the allowed hosts list and the NSClient++ service was restarted.

6. Verify NRPE connectivity: NRPE connectivity from the Nagios container to the Windows host was verified successfully.

7. Add Windows host: Windows_Host_001 was added to Nagios and services for CPU Load, Memory Usage and C Drive Space were configured using the check_nrpe command.

8. Validate Nagios configuration: The Nagios pre-flight configuration check completed with 0 errors.

9. Monitor the Windows host: The Nagios Current Status → Services page was used to verify the live monitoring results.

Important Commands / Outputs

NRPE connectivity

```
docker exec nagios4 /opt/nagios/libexec/check_nrpe -H host.docker.internal
I (0.6.13 2025-03-14) seem to be doing fine...
```

CPU check

```
docker exec nagios4 /opt/nagios/libexec/check_nrpe -H host.docker.internal -c check_cpu
OK: CPU load is ok.|'total 5m'=23%;80;90 'total 1m'=16%;80;90 'total 5s'=13%;80;90
```

C Drive check

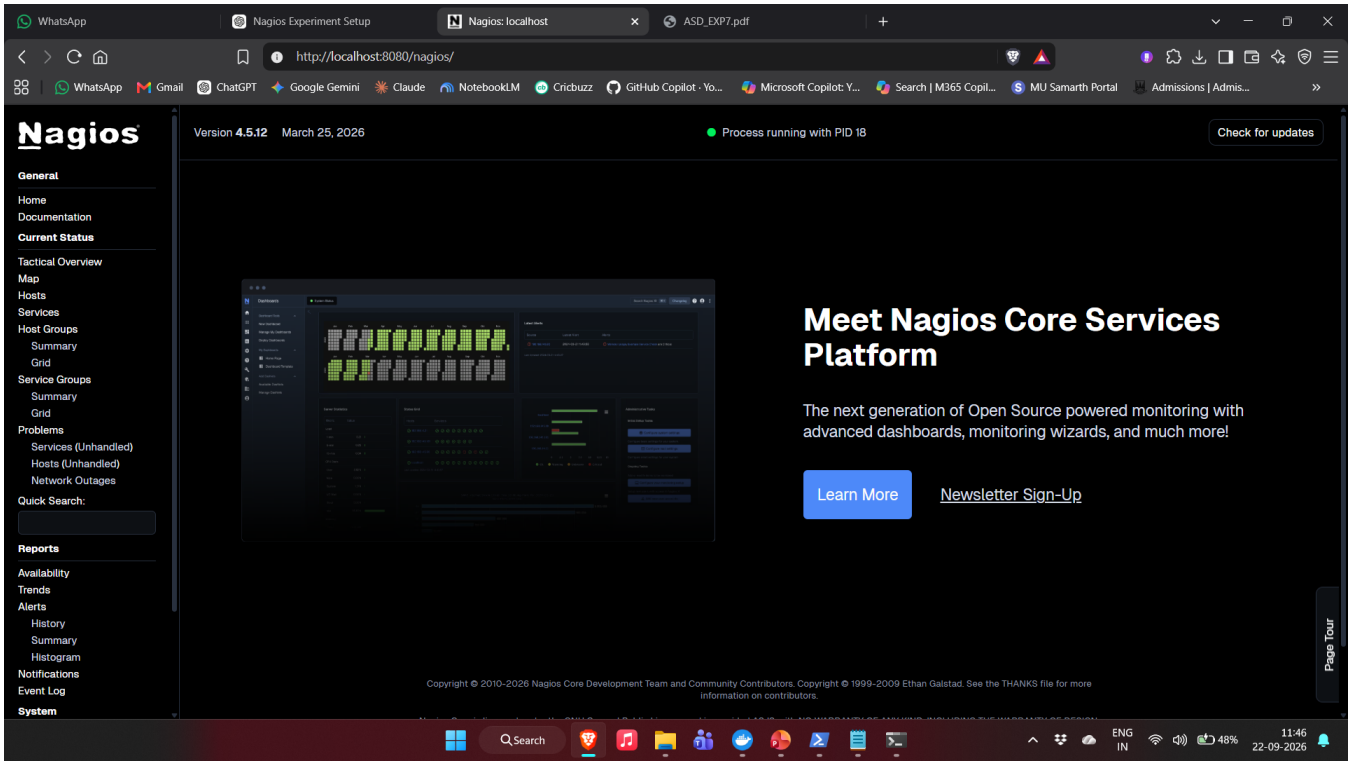
```
check_drivesize → OK All 1 drive(s) are ok
```

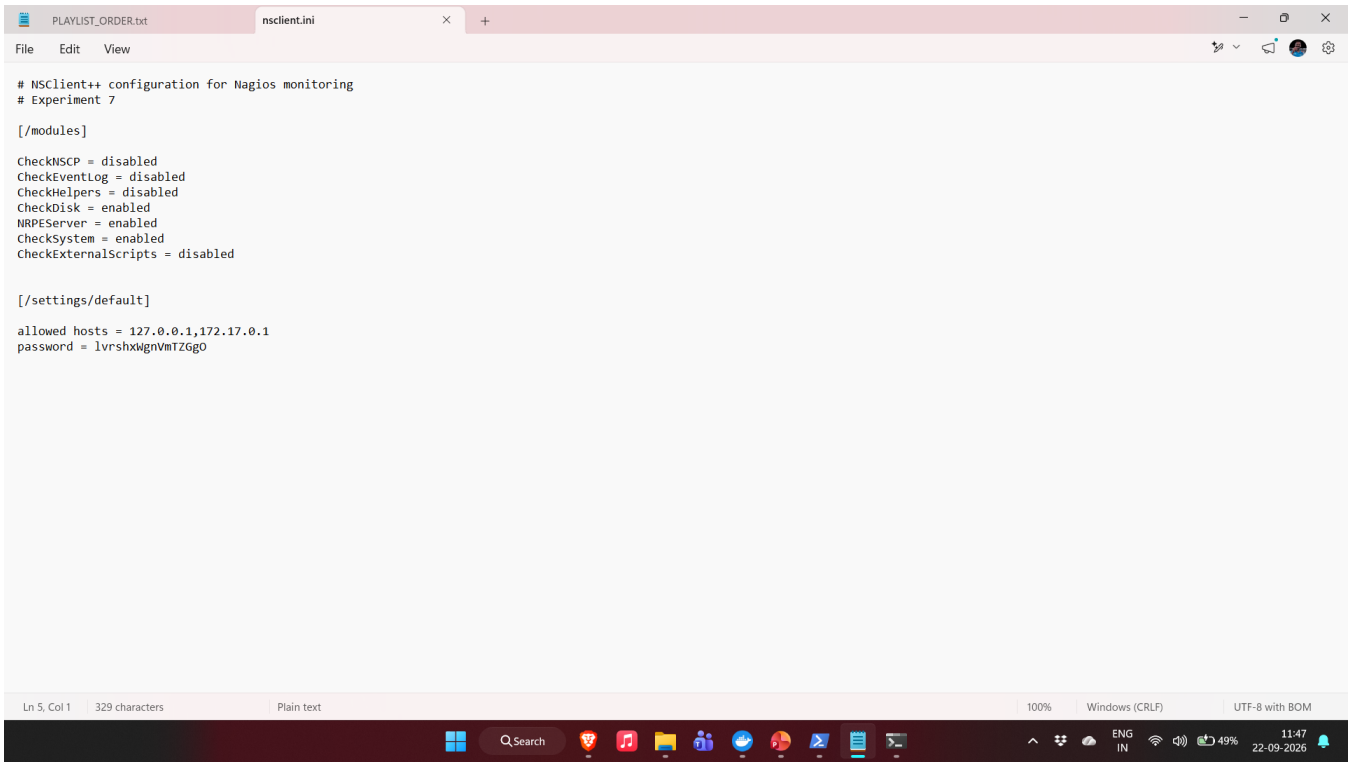
Memory check

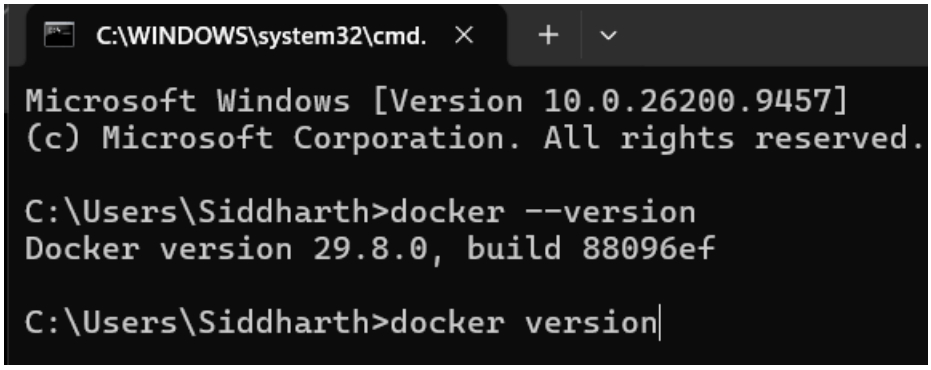
```
check_memory → WARNING: committed = 18.784GB, physical = 12.341GB
```

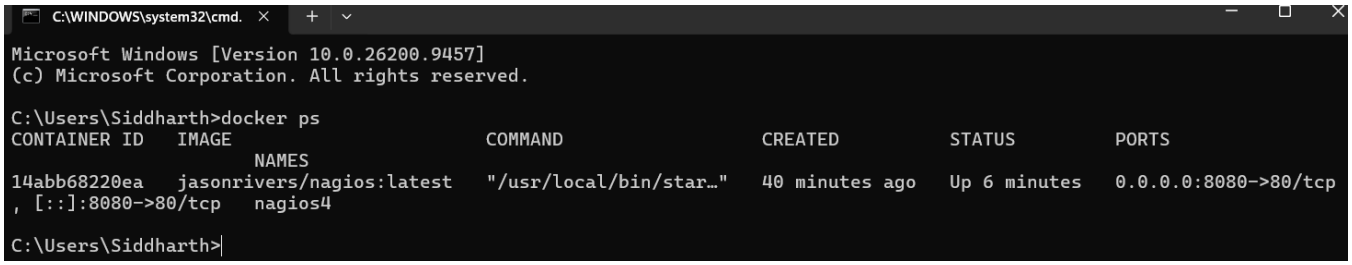
Nagios configuration validation

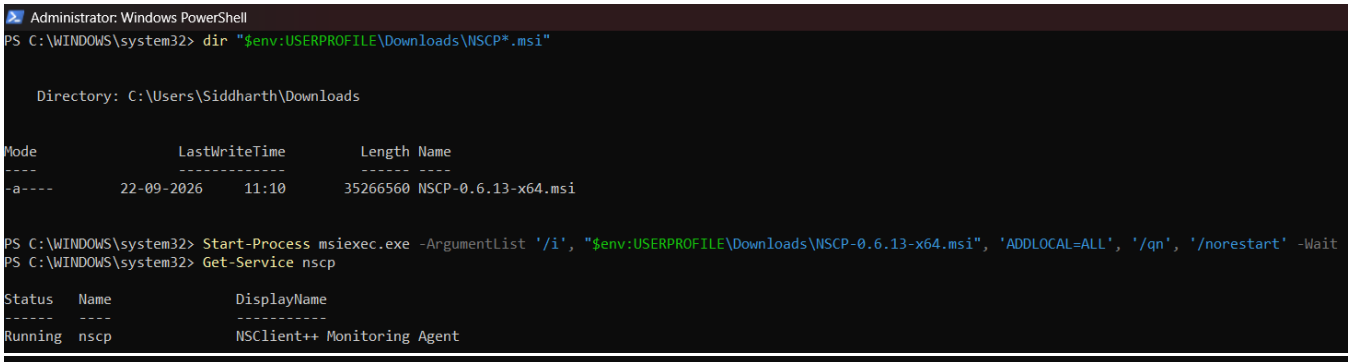
```
Total Warnings: 7
Total Errors:   0
Things look okay - No serious problems were detected during the pre-flight check
```

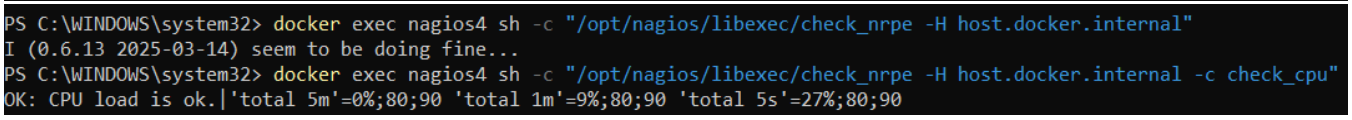

WhatsApp
Nagios Experiment Setup
Nagios: localhost
ASD_EXP7.pdf
http://localhost:8080/nagios/
Nagios
Version 4.5.12 March 25, 2026
Process running with PID 18
Check for updates
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Meet Nagios Core Services Platform
The next generation of Open Source powered monitoring with advanced dashboards, monitoring wizards, and much more!
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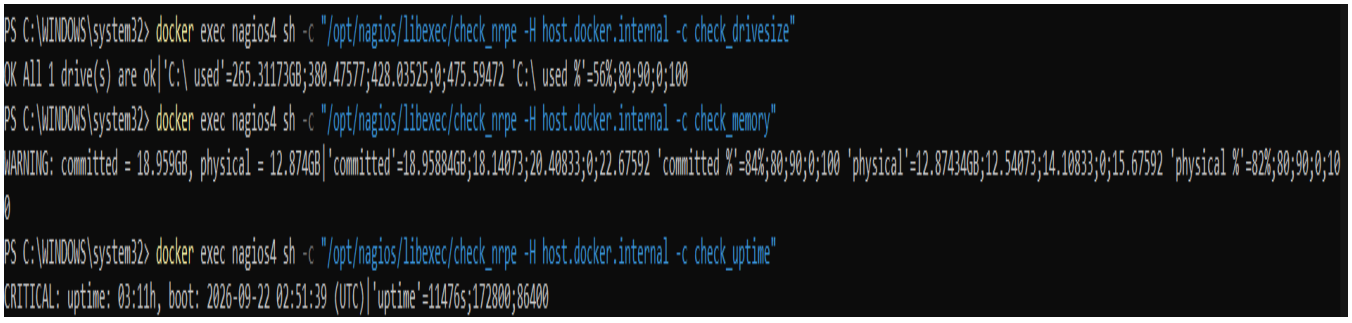

PLAYLIST_ORDER.txt
nsclient.ini
File Edit View
NSClient++ configuration for Nagios monitoring
Experiment 7
[/modules]
CheckNSCP = disabled
CheckEventLog = disabled
CheckHelpers = disabled
CheckDisk = enabled
NRPEServer = enabled
CheckSystem = enabled
CheckExternalScripts = disabled
[/settings/default]
allowed hosts = 127.0.0.1,172.17.0.1
password = lvrshxWgnVmTZGgO
Ln 5, Col 1
329 characters
Plain text
100%
Windows (CRLF)
UTF-8 with BOM


C:\WINDOWS\system32\cmd.
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Siddharth>docker --version
Docker version 29.8.0, build 88096ef
C:\Users\Siddharth>docker version


C:\WINDOWS\system32\cmd.
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Siddharth>docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
14abb68220ea jasonrivers/nagios:latest "/usr/local/bin/star…" 40 minutes ago Up 6 minutes 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp nagios4
C:\Users\Siddharth>


Administrator: Windows PowerShell
PS C:\WINDOWS\system32> dir "$env:USERPROFILE\Downloads\NSCP*.msi"
Directory: C:\Users\Siddharth\Downloads
Mode LastWriteTime Length Name
-a---- 22-09-2026 11:10 35266560 NSCP-0.6.13-x64.msi
PS C:\WINDOWS\system32> Start-Process msiexec.exe -ArgumentList '/i', "$env:USERPROFILE\Downloads\NSCP-0.6.13-x64.msi", 'ADDLOCAL=ALL', '/qn', '/norestart' -Wait
PS C:\WINDOWS\system32> Get-Service nscp
Status Name DisplayName
Running nscp NSClient++ Monitoring Agent


PS C:\WINDOWS\system32> docker exec nagios4 sh -c "/opt/nagios/libexec/check_nrpe -H host.docker.internal"
I (0.6.13 2025-03-14) seem to be doing fine...
PS C:\WINDOWS\system32> docker exec nagios4 sh -c "/opt/nagios/libexec/check_nrpe -H host.docker.internal -c check_cpu"
OK: CPU load is ok.|'total 5m'=0%;80;90 'total 1m'=9%;80;90 'total 5s'=27%;80;90


PS C:\WINDOWS\system32> docker exec nagios4 sh -c "/opt/nagios/libexec/check_nrpe -H host.docker.internal -c check_drivesize"
OK All 1 drive(s) are ok|'C:\ used'=265.31173GB;380.47577;428.03525;0;475.59472 'C:\ used %'=56%;80;90;0;100
PS C:\WINDOWS\system32> docker exec nagios4 sh -c "/opt/nagios/libexec/check_nrpe -H host.docker.internal -c check_memory"
WARNING: committed = 18.959GB, physical = 12.874GB|'committed'=18.95884GB;18.14073;20.40833;0;22.67592 'committed %'=84%;80;90;0;100 'physical'=12.87434GB;12.54073;14.10833;0;15.67592 'physical %'=82%;80;90;0;100
PS C:\WINDOWS\system32> docker exec nagios4 sh -c "/opt/nagios/libexec/check_nrpe -H host.docker.internal -c check_uptime"
CRITICAL: uptime: 03:11h, boot: 2026-09-22 02:51:39 (UTC)|'uptime'=11476s;172800;86400

```
PS C:\WINDOWS\system32> docker exec nagios4 /opt/nagios/bin/nagios -v /opt/nagios/etc/nagios.cfg

Nagios Core 4.5.12
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2026-03-25
License: GPL

Website: https://www.nagios.org
Reading configuration data...
   Read main config file okay...
   Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
Warning: Service 'C Drive Space' on host 'Windows_Host_001' has no default contacts or contactgroups defined!
Warning: Service 'C Drive Space' on host 'Windows_Host_001' has no notification time period defined!
Warning: Service 'CPU Load' on host 'Windows_Host_001' has no default contacts or contactgroups defined!
Warning: Service 'CPU Load' on host 'Windows_Host_001' has no notification time period defined!
Warning: Service 'Memory Usage' on host 'Windows_Host_001' has no default contacts or contactgroups defined!
Warning: Service 'Memory Usage' on host 'Windows_Host_001' has no notification time period defined!
        Checked 10 services.
Warning: Host 'Windows_Host_001' has no default contacts or contactgroups defined!
        Checked 2 hosts.
        Checked 1 host groups.
        Checked 0 service groups.
        Checked 1 contacts.
        Checked 1 contact groups.
        Checked 24 commands.
        Checked 5 time periods.
        Checked 0 host escalations.
        Checked 0 service escalations.
Checking for circular paths...
        Checked 2 hosts
        Checked 0 service dependencies
        Checked 0 host dependencies
        Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 7
Total Errors:   0

Things look okay - No serious problems were detected during the pre-flight check
```

Nagios Monitoring Configuration

Host	Service	Command
Windows_Host_001	CPU Load	check_nrpe!check_cpu
Windows_Host_001	Memory Usage	check_nrpe!check_memory
Windows_Host_001	C Drive Space	check_nrpe!check_drivesize

Result / Observation

Nagios successfully monitored the Windows host through NSClient++ and NRPE. The CPU Load and C Drive Space services were reported as OK. Memory Usage was reported as WARNING because the measured memory usage crossed the configured threshold. The monitoring services were actively checked and no pending checks remained.

Final Output – Nagios Current Status

The final Nagios dashboard showed the following status for Windows_Host_001:

- C Drive Space — OK
- CPU Load — OK
- Memory Usage — WARNING
- Pending checks — 0

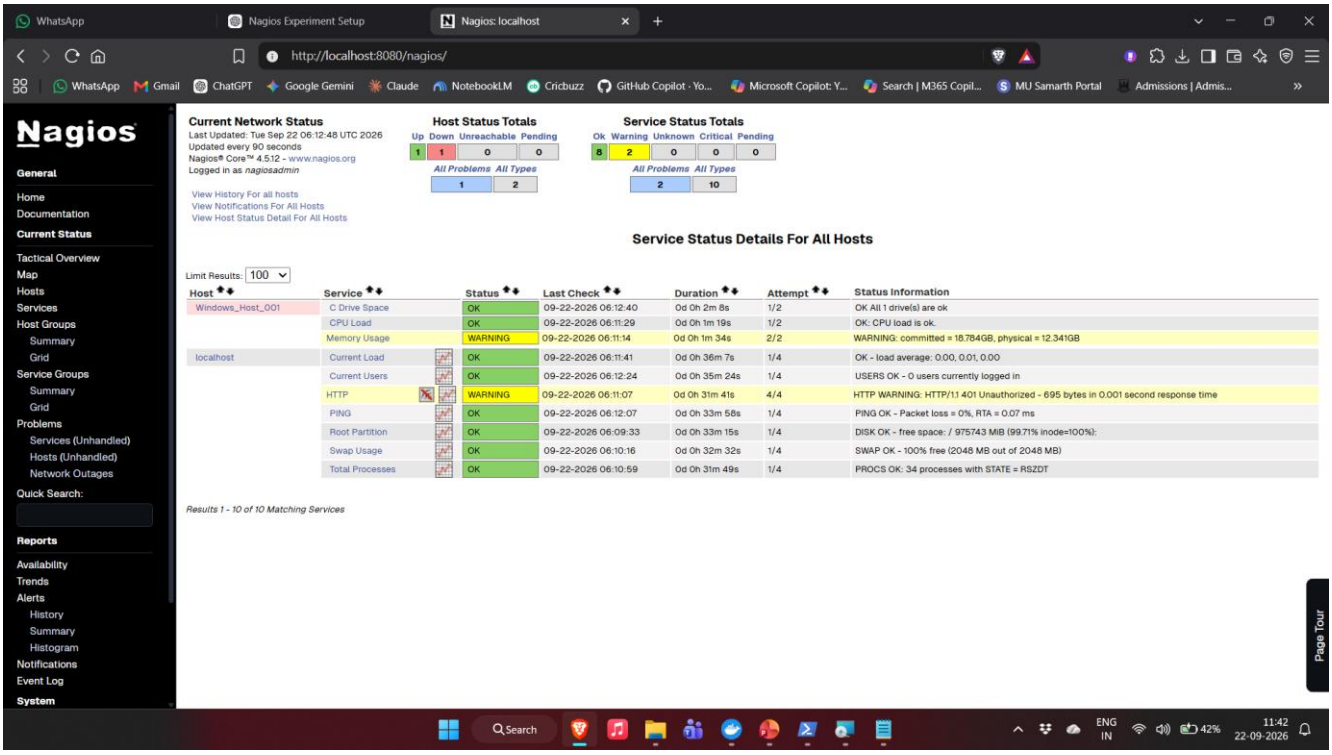


Figure: Final Nagios Current Status → Services screen

Conclusion / Findings

The Nagios monitoring system was successfully implemented using Docker and integrated with a Windows host using NSClient++ and NRPE. CPU load, memory usage and C: drive space were successfully monitored through the Nagios web interface. The monitoring dashboard displayed the real-time status of the configured Windows host and its services.